

PSTAT 100 Data Science Concepts and Analysis

Assignment 3

Author: Date:

Submission Instructions

This assignment will be **due for submission on Wednesday May 27th at 11:59PM**. Please ensure you submit a `.pdf` file to Canvas by the posted time, no other file types will be accepted. Please see the course website for some information regarding exporting Jupyter notebooks to pdf, and if you are having exporting problems discuss them with a member of the teaching staff in their office hours.

Collaboration

You are encouraged to work on your assignments independently. Please only collaborate with others should you require assistance with more challenging problems. Should you need to collaborate with others, please note their names here:

Collaborators: None

Agent Usage

Additionally, you are permitted to use resources such as ChatGPT and Claude to help you with your homework assignments and to enhance your learning experience. Please make a note of any agents you have used in this submission here:

Agents: Claude

WARNING: Please ensure you check and understand the outputs these agents produce. Should your solution show evidence of abuse of AI usage (i.e. use external data not detailed in this assignment, use methods not covered in class) your solution will receive a score of 0.

1. Introduction

In this report we will be performing a statistical analysis to investigate the effect of economic, demographic and health factors on human life expectancy across various

countries. Our study will be utilizing the [Life Expectancy \(WHO\)](#) data, made available on Kaggle by Kumar Rajarshi and originally collected from WHO and United Nations website with the help of Deeksha Russell and Duan Wang.

In this project we consider data from year 2000-2015 for 193 countries for further analysis. This will involve:

- **Data Preparation:**
 - Data Importing
 - Missingness.
 - Data Transformations.
- **Exploratory Data Analysis:**
 - Summary Statistic Tables
 - Univariate Distributions
 - Bivariate Relationships
- **Modelling:**
 - Fit a Simple Linear Regression (SLR) model.
 - Fit several Multiple Linear Regression (MLR) models.
 - Evaluate model performance.
 - Residual analysis.

In this report we will be making use of the following libraries:

```
In [1]: # Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import statsmodels.formula.api as smf
from scipy.stats import probplot

# Figure formatting
sns.set_style("whitegrid")
sns.set_palette("Set2")
```

2. Data Preparation

In this section we will load and prepare the data. Download the

`life_expectancy.csv` file from Canvas. Print the first 5 rows of this data set as well as the shape of the data using the chunk below:

```
In [2]: # Load data
df = pd.read_csv("/Users/cyrusisaacs/PSTAT 100/Data/life_expectancy.csv")
# Print shape
print("Shape: ", df.shape)
```

```
# Print column names
print(f"  Columns ({len(df.columns)}):")
for c in df.columns:
    print(f"    {c!r}")
# Print first 5 rows
df.head()
```

```
Shape: (2938, 22)
Columns (22):
  'Country'
  'Year'
  'Status'
  'Life expectancy '
  'Adult Mortality'
  'infant deaths'
  'Alcohol'
  'percentage expenditure'
  'Hepatitis B'
  'Measles '
  ' BMI '
  'under-five deaths '
  'Polio'
  'Total expenditure'
  'Diphtheria '
  ' HIV/AIDS'
  'GDP'
  'Population'
  ' thinness 1-19 years'
  ' thinness 5-9 years'
  'Income composition of resources'
  'Schooling'
```

Out[2]:

	Country	Year	Status	Life expectancy	Adult Mortality	infant deaths	Alcohol	percentage expenditure
0	Afghanistan	2015	Developing	65.0	263.0	62	0.01	71.279624
1	Afghanistan	2014	Developing	59.9	271.0	64	0.01	73.523582
2	Afghanistan	2013	Developing	59.9	268.0	66	0.01	73.219243
3	Afghanistan	2012	Developing	59.5	272.0	69	0.01	78.184215
4	Afghanistan	2011	Developing	59.2	275.0	71	0.01	7.097109

5 rows x 22 columns

We notice that the data column names are inconsistently formatted which will make our life more challenging in future if we do not address it. To see

!! Problem

Q2.1 Column Formatting

Reformat the column names in the following ways:

1. Remove whitespace (spaces before and after) using `strip`.
2. Remove all capitalization.
3. Replace internal spacing with underscores "_".
4. Replace any character that is **not** a lowercase letter, digit or underscore.

Hint: For (4) look up what the regex `[^a-z0-9_]` means and how it can be used with `str.replace(...)`.

In [3]: *# Your solution here*

```
df.columns = (  
    df.columns  
    .str.strip()  
    .str.lower()  
    .str.replace(' ', '_')  
    .str.replace(r'[^a-z0-9_]', '', regex=True)  
)
```

Use the following chunk to ensure that our column relabelling was successful. You should see 22 column names with only lower case letters, numbers and underscores:

In [4]: *# Print cleaned column names*

```
print(f" Columns ({len(df.columns)}):")  
for c in df.columns:  
    print(f" {c!r}")
```

```
Columns (22):
  'country'
  'year'
  'status'
  'life_expectancy'
  'adult_mortality'
  'infant_deaths'
  'alcohol'
  'percentage_expenditure'
  'hepatitis_b'
  'measles'
  'bmi'
  'underfive_deaths'
  'polio'
  'total_expenditure'
  'diphtheria'
  'hivaids'
  'gdp'
  'population'
  'thinness__119_years'
  'thinness_59_years'
  'income_composition_of_resources'
  'schooling'
```

Next we inspect the data types used in each column using the chunk below:

```
In [5]: # Check data types
df.dtypes
```

```
Out[5]: country          str
year          int64
status        str
life_expectancy  float64
adult_mortality  float64
infant_deaths   int64
alcohol         float64
percentage_expenditure  float64
hepatitis_b     float64
measles         int64
bmi            float64
underfive_deaths  int64
polio          float64
total_expenditure  float64
diphtheria      float64
hivaids        float64
gdp            float64
population      float64
thinness__119_years  float64
thinness_59_years  float64
income_composition_of_resources  float64
schooling       float64
dtype: object
```

All we need to do is convert `status` into a category type which we do as follows:

```
In [6]: # Change status to categorical variable
df["status"] = df["status"].astype("category")
```

Next we proceed to evaluating the missingness of the data.

!! Problem

Q2.2 Missingness Tables

Produce the following tables:

- A standard missingness table showing the counts and percentages of missing data for each column.
- A missingness table showing the percentages of missing data for each column, grouped by status.

We will be using simple imputation (replacement with median) to handle missing values in numerical columns. Using this table and by reading the data documentation on Kaggle, comment on the validity of this choice. Consider the following questions:

- Is the data Missing Completely At Random (MCAR)?
- Might there be underlying patterns to the missingness?

```
In [7]: missing_table = pd.DataFrame({
    'Missing Count': df.isnull().sum(),
    'Missing Percent': (df.isnull().mean() * 100)
})

missing_table = missing_table.sort_values(
    by='Missing Percent',
    ascending=False
)

print("Standard Missingness Table")
print(missing_table)

missing_by_status = (
    df.groupby('status')
    .apply(lambda x: x.isnull().mean() * 100)
    .T
)

print("\nMissingness Percentage by Status")
print(missing_by_status)
```

Standard Missingness Table

	Missing Count	Missing Percent
population	652	22.191967
hepatitis_b	553	18.822328
gdp	448	15.248468
total_expenditure	226	7.692308
alcohol	194	6.603131
income_composition_of_resources	167	5.684139
schooling	163	5.547992
thinness_59_years	34	1.157250
thinness__119_years	34	1.157250
bmi	34	1.157250
polio	19	0.646698
diphtheria	19	0.646698
life_expectancy	10	0.340368
adult_mortality	10	0.340368
hivaids	0	0.000000
country	0	0.000000
year	0	0.000000
measles	0	0.000000
percentage_expenditure	0	0.000000
infant_deaths	0	0.000000
status	0	0.000000
underfive_deaths	0	0.000000

Missingness Percentage by Status

	Developed	Developing
status		
country	0.000000	0.000000
year	0.000000	0.000000
life_expectancy	0.000000	0.412201
adult_mortality	0.000000	0.412201
infant_deaths	0.000000	0.000000
alcohol	5.468750	6.842539
percentage_expenditure	0.000000	0.000000
hepatitis_b	33.789062	15.663644
measles	0.000000	0.000000
bmi	0.000000	1.401484
underfive_deaths	0.000000	0.000000
polio	0.000000	0.783182
total_expenditure	6.250000	7.996702
diphtheria	0.000000	0.783182
hivaids	0.000000	0.000000
gdp	12.500000	15.828524
population	18.750000	22.918384
thinness__119_years	0.000000	1.401484
thinness_59_years	0.000000	1.401484
income_composition_of_resources	9.375000	4.905194
schooling	9.375000	4.740313

Is the data MCAR? No. The data is unlikely to be Missing Completely At Random (MCAR), since missingness probably depends on factors such as country development level, healthcare infrastructure, and reporting quality.

Might there be underlying patterns to the missingness? Yes. Developing countries may have more missing data for variables like GDP, healthcare spending, and vaccinations, suggesting systematic patterns in the missingness.

!! Problem

Q2.3 Missing Data Imputation

First, isolate the 20 columns that are numeric and store their names in `numeric_cols`. Then, write a `for` loop that iterates over these numeric columns and performs the following:

1. Replaces any missing values with the median of that column for that country.
2. Replaces any remaining missing values with the median for that entire column.

Check that your imputation has successfully removed all missing values by using the following:

```
print(f"\n After imputation – remaining nulls:
{df.isnull().sum().sum()}")
```

```
In [8]: numeric_cols = df.select_dtypes(include='number').columns

for col in numeric_cols:

    df[col] = df.groupby('country')[col].transform(
        lambda x: x.fillna(x.median())
    )

    df[col] = df[col].fillna(df[col].median())

print(f"\nAfter imputation – remaining nulls: {df.isnull().sum().sum()}")
```

After imputation – remaining nulls: 0

We conclude our preparation by noting that several covariates are very heavily right skewed. We can see this by producing a multiplot of the histograms of all numerical variables:

```
In [9]: fig, axes = plt.subplots(5, 4, figsize=(16, 15))
axes = axes.flatten()

for i, col in enumerate(numeric_cols):
    sns.histplot(df[col].dropna(), ax=axes[i], bins=30, color="steelblue")
```

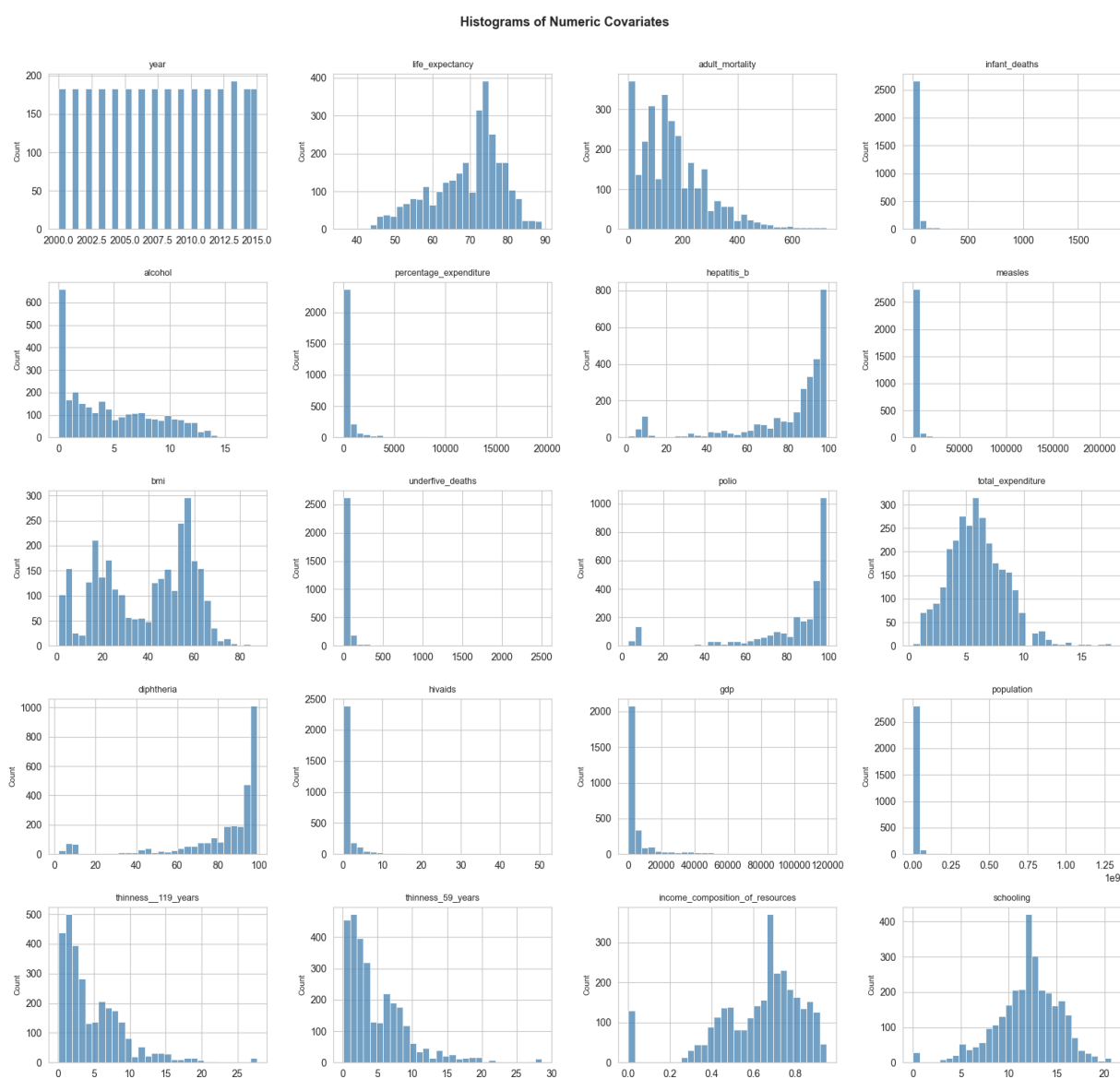
```

axes[i].set_title(col, fontsize=9)
axes[i].set_xlabel("")
axes[i].set_ylabel("Count", fontsize=8)

# Hide any unused axes
for j in range(len(numeric_cols), len(axes)):
    axes[j].set_visible(False)

plt.suptitle("Histograms of Numeric Covariates", fontsize=13, fontweight="bold")
plt.tight_layout()
plt.show()

```



We therefore use the following chunk to take the log of these variables.

```

In [10]: skewed = ['infant_deaths', 'percentage_expenditure', 'measles', 'underfive_c

for col in skewed:
    if col in df.columns:
        df[col] = np.log1p(df[col].clip(lower=0))

```

3. Exploratory Data Analysis

3.1 Summary Statistics

We proceed to our data exploration and start by producing tables of summary statistics.

!! Problem

Q3.1 Summary Statistics

Produce a table of the key summary statistics (mean, standard deviation, minimum, LQ, median, UQ, max, range, skew and kurtosis) for all numerical covariates. Provide a few comments on the results including observations about skewed covariates, potential outliers or other points of note.

```
In [11]: summary_table = pd.DataFrame({
    'Mean': df[numeric_cols].mean(),
    'Std Dev': df[numeric_cols].std(),
    'Min': df[numeric_cols].min(),
    'LQ': df[numeric_cols].quantile(0.25),
    'Median': df[numeric_cols].median(),
    'UQ': df[numeric_cols].quantile(0.75),
    'Max': df[numeric_cols].max(),
    'Range': df[numeric_cols].max() - df[numeric_cols].min(),
    'Skew': df[numeric_cols].skew(),
    'Kurtosis': df[numeric_cols].kurtosis()
})

summary_table = summary_table.round(3)

print(summary_table)
```

	Mean	Std Dev	Min	LQ	\
year	2007.519	4.614	2000.000	2004.000	
life_expectancy	69.235	9.509	36.300	63.200	
adult_mortality	164.726	124.086	1.000	74.000	
infant_deaths	1.769	1.663	0.000	0.000	
alcohol	4.605	4.034	0.010	0.923	
percentage_expenditure	3.986	2.730	0.000	1.738	
hepatitis_b	80.100	24.554	1.000	75.000	
measles	3.357	3.259	0.000	0.000	
bmi	38.381	19.935	1.000	19.400	
underfive_deaths	1.940	1.782	0.000	0.000	
polio	82.417	23.478	3.000	78.000	
total_expenditure	5.917	2.469	0.370	4.260	
diphtheria	82.197	23.757	2.000	78.000	
hivaids	0.510	0.766	0.095	0.095	
gdp	7.458	1.728	0.986	6.333	
population	13.950	2.349	3.555	12.945	
thinness__119_years	4.822	4.398	0.100	1.600	
thinness_59_years	4.852	4.486	0.100	1.600	
income_composition_of_resources	0.630	0.205	0.000	0.504	
schooling	12.010	3.265	0.000	10.300	

	Median	UQ	Max	Range	Ske
w \					
year	2008.000	2012.000	2015.000	15.000	-0.00
6					
life_expectancy	72.100	75.600	89.000	52.700	-0.64
2					
adult_mortality	144.000	227.000	723.000	722.000	1.17
8					
infant_deaths	1.386	3.135	7.496	7.496	0.70
2					
alcohol	3.770	7.680	17.870	17.860	0.59
2					
percentage_expenditure	4.188	6.093	9.877	9.877	-0.07
8					
hepatitis_b	91.000	96.000	99.000	98.000	-1.82
9					
measles	2.890	5.890	12.265	12.265	0.55
5					
bmi	43.500	56.100	87.300	86.300	-0.22
9					
underfive_deaths	1.609	3.367	7.824	7.824	0.64
7					
polio	93.000	97.000	99.000	96.000	-2.07
1					
total_expenditure	5.730	7.458	17.600	17.230	0.62
8					
diphtheria	93.000	97.000	99.000	97.000	-2.04
7					
hivaids	0.095	0.588	3.944	3.848	2.12
7					
gdp	7.455	8.472	11.688	10.702	-0.13
5					
population	14.149	15.338	20.981	17.426	-0.70
3					

thinness__119_years 9	3.300	7.100	27.700	27.600	1.72
thinness_59_years 6	3.300	7.200	28.600	28.500	1.79
income_composition_of_resources 3	0.677	0.772	0.948	0.948	-1.21
schooling 5	12.300	14.100	20.700	20.700	-0.63

	Kurtosis
year	-1.214
life_expectancy	-0.224
adult_mortality	1.767
infant_deaths	-0.329
alcohol	-0.796
percentage_expenditure	-1.021
hepatitis_b	2.469
measles	-0.883
bmi	-1.271
underfive_deaths	-0.507
polio	3.656
total_expenditure	1.212
diphtheria	3.451
hivaids	4.079
gdp	0.089
population	1.384
thinness__119_years	4.060
thinness_59_years	4.453
income_composition_of_resources	1.694
schooling	1.124

Several variables such as gdp, population, measles, and percentage_expenditure were strongly right-skewed before transformation. The log transformation reduced this skewness substantially. Some variables may still show moderate skewness and high kurtosis, indicating the presence of extreme values or outliers. Variables with high kurtosis suggest heavier tails than a normal distribution, meaning a small number of countries have unusually extreme values.

3.2 Univariate Analysis

First, lets consider our categorical covariate `status`.

!! Problem

Q3.2 Countplot of `status`

Produce a countplot of the covariate `status` to visualize the proportion of developing countries against developed countries. Comment on your resulting plot.

```
In [12]: plt.figure(figsize=(6, 4))

sns.countplot(
    data=df,
    x='status'
)

plt.title('Distribution of Country Status')
plt.xlabel('Status')
plt.ylabel('Count')

plt.show()
```



The dataset is heavily dominated by developing countries, with substantially fewer developed countries represented. This indicates a class imbalance in the status variable. The imbalance reflects the real world distribution of countries included in the WHO dataset, where more observations are available from developing nations across multiple years. This imbalance should be considered in later analyses, since relationships in the data may be influenced more strongly by patterns within developing countries.

!! Problem

Q3.3 Histogram Multiplot

Produce a multiplot with 5 rows and 4 columns showing estimates for the distributions for the 20 numerical covariates. Discuss your findings and the distributions of our

transformed covariates.

```
In [13]: fig, axes = plt.subplots(5, 4, figsize=(20, 20))

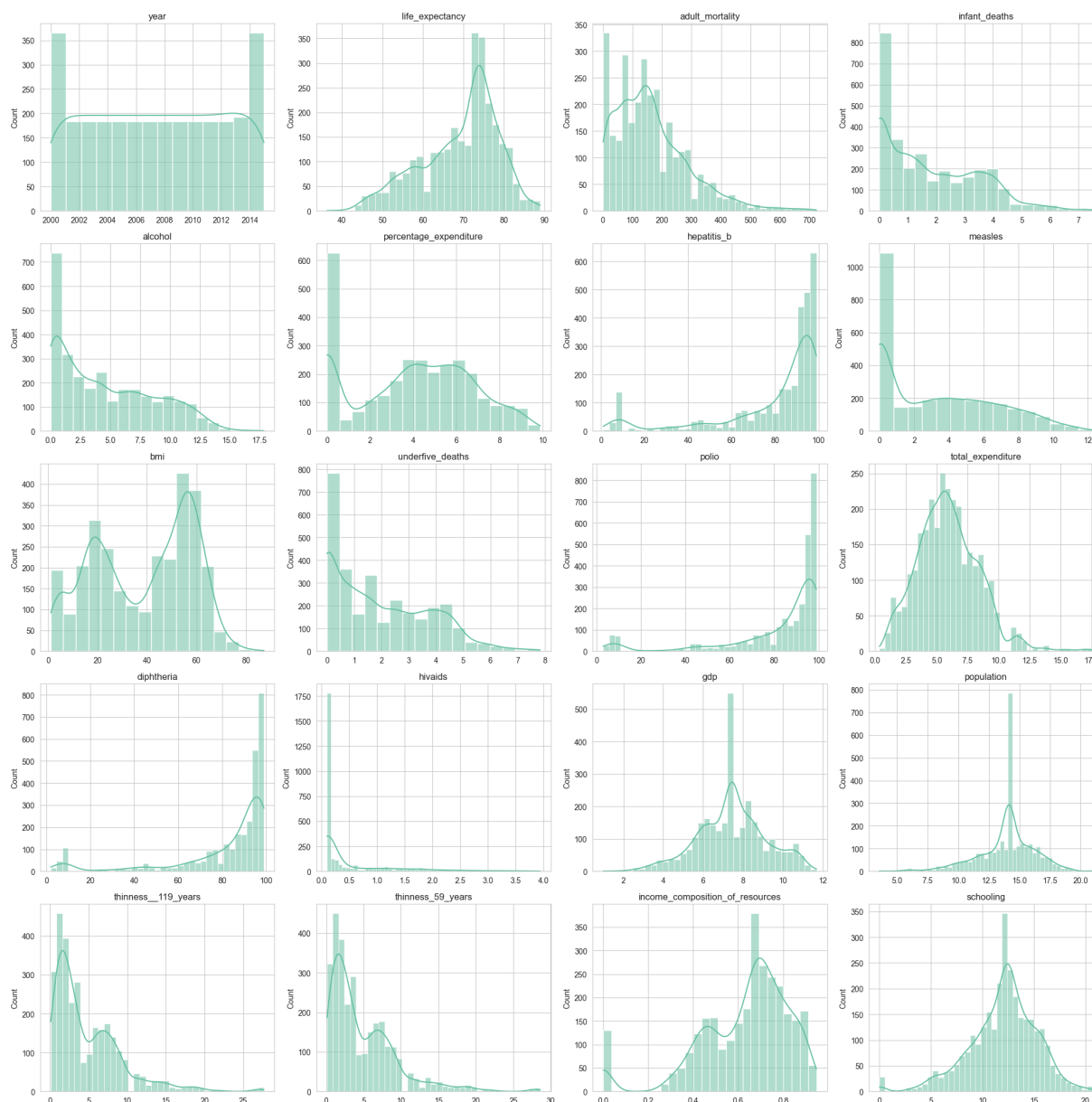
axes = axes.flatten()

for i, col in enumerate(numeric_cols):

    sns.histplot(
        df[col],
        kde=True,
        ax=axes[i]
    )

    axes[i].set_title(col)
    axes[i].set_xlabel('')
    axes[i].set_ylabel('Count')

plt.tight_layout()
plt.show()
```



Many of the numerical covariates originally showed strong right skewness, particularly: gdp, population, percentage_expenditure, measles, underfive_deaths, infant_deaths. After applying the log transformation, these variables appear substantially more symmetric and less extreme. The transformed distributions still show some skewness, but the long right tails have been reduced considerably. Several variables such as vaccination rates remain concentrated near high values, producing left-skewed distributions. Variables related to mortality and disease burden still display wider spread and occasional extreme observations, reflecting large differences between countries. Some distributions appear approximately bell-shaped after transformation, making them more suitable for statistical modelling techniques that assume moderate normality.

3.3 Bivariate Analysis

We begin our bivariate analysis by looking at the relationship between our response and the categorical variable `status`.

!! Problem

Q3.4 Violin and Grouped Histogram Plot

Produce a multiplot containing the following:

1. A violin plot of `life_expectancy` against `status`; and
2. A histogram of `life_expectancy` grouped by `status`.

Comment on whether you see evidence of a significant difference in the distribution of `life_expectancy` between developed and developing countries.

```
In [14]: fig, axes = plt.subplots(1, 2, figsize=(14, 5))

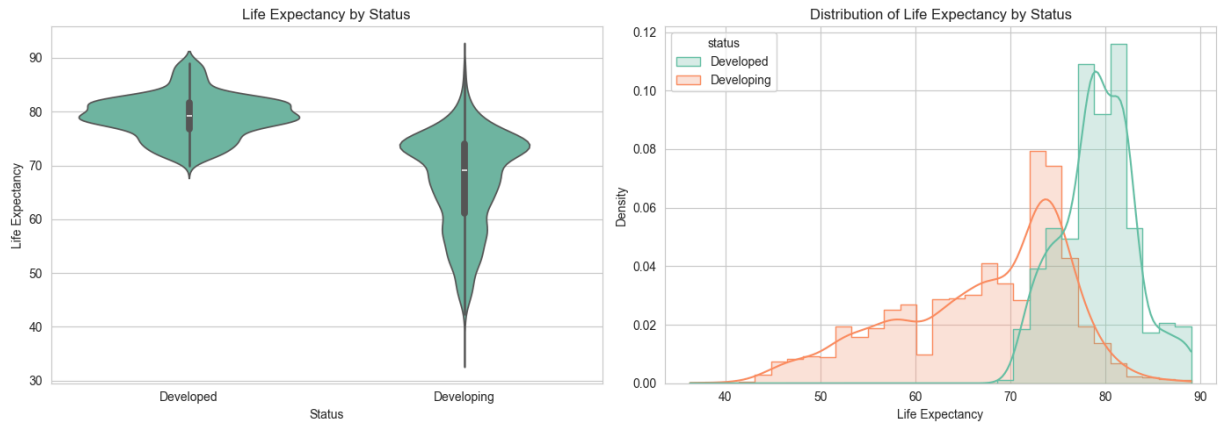
sns.violinplot(
    data=df,
    x='status',
    y='life_expectancy',
    ax=axes[0]
)

axes[0].set_title('Life Expectancy by Status')
axes[0].set_xlabel('Status')
axes[0].set_ylabel('Life Expectancy')

sns.histplot(
    data=df,
    x='life_expectancy',
    hue='status',
    kde=True,
    element='step',
    stat='density',
    common_norm=False,
    ax=axes[1]
)

axes[1].set_title('Distribution of Life Expectancy by Status')
axes[1].set_xlabel('Life Expectancy')
axes[1].set_ylabel('Density')

plt.tight_layout()
plt.show()
```



Developed countries generally have substantially higher life expectancy than developing countries. The violin plot shows that developed countries have distributions concentrated at higher life expectancy values with less variability. Developing countries show a wider spread and lower median life expectancy, indicating greater variation in health outcomes. The grouped histogram also shows a clear shift: developed countries are concentrated toward higher life expectancy values, while developing countries are more spread out and include many lower values. Overall, the plots provide strong visual evidence of a significant difference in life expectancy distributions between developed and developing countries.

!! Problem

Q3.5 Correlation Heat Map

Produce a correlation heat map of the numerical variables in the dataset. Write some code to return the covariates with the strongest correlations. Comment on the strongest positive and negative correlations you observe.

```
In [15]: corr_matrix = df[numeric_cols].corr()
plt.figure(figsize=(16, 12))
sns.heatmap(
    corr_matrix,
    cmap='coolwarm',
    center=0,
    annot=False
)

plt.title('Correlation Heatmap of Numerical Variables')
plt.show()

corr_pairs = corr_matrix.unstack()

corr_pairs = corr_pairs[
    corr_pairs.index.get_level_values(0)
```

```

    != corr_pairs.index.get_level_values(1)
]

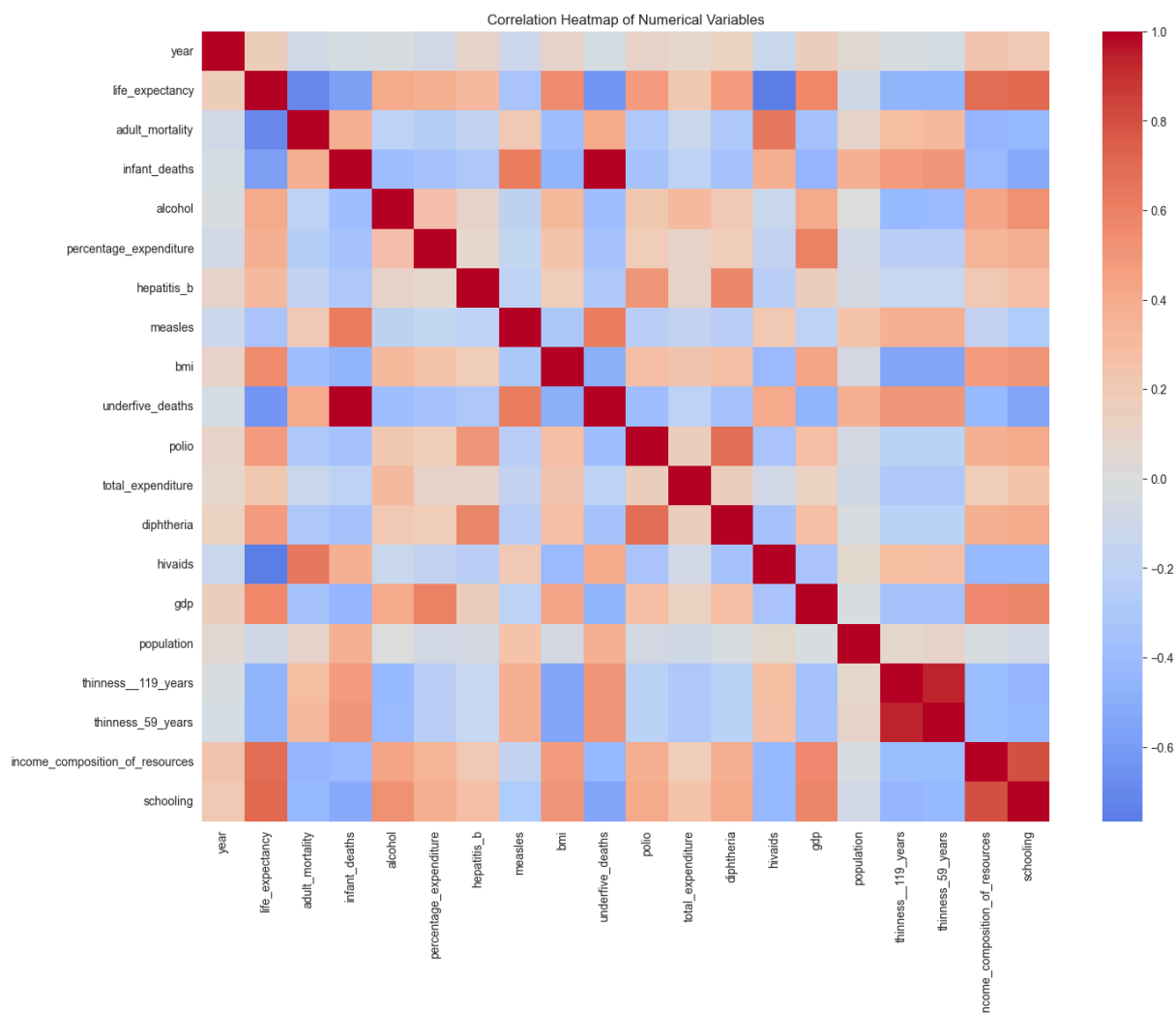
corr_pairs = corr_pairs.drop_duplicates()

sorted_corr = corr_pairs.sort_values()

print("Strongest Negative Correlations")
print(sorted_corr.head(10))

print("\nStrongest Positive Correlations")
print(sorted_corr.tail(10))

```



Strongest Negative Correlations

life_expectancy	hivaids	-0.766335
	adult_mortality	-0.696390
	underfive_deaths	-0.621177
	infant_deaths	-0.592058
underfive_deaths	schooling	-0.541438
bmi	thinness_59_years	-0.539365
	thinness__119_years	-0.532487
infant_deaths	schooling	-0.521825
bmi	underfive_deaths	-0.489748
infant_deaths	bmi	-0.473661

dtype: float64

Strongest Positive Correlations

percentage_expenditure	gdp	0.602730
infant_deaths	measles	0.623712
measles	underfive_deaths	0.623848
adult_mortality	hivaids	0.635194
polio	diphtheria	0.676887
life_expectancy	income_composition_of_resources	0.688662
	schooling	0.713054
income_composition_of_resources	schooling	0.795383
thinness__119_years	thinness_59_years	0.939187
infant_deaths	underfive_deaths	0.996315

dtype: float64

Strong positive correlations are observed between: life_expectancy and schooling
 life_expectancy and income_composition_of_resources vaccination-related variables
 such as polio and diphtheria These relationships suggest that higher education, stronger
 economic development, and better healthcare systems are associated with longer life
 expectancy. Strong negative correlations are commonly observed between:
 life_expectancy and adult_mortality life_expectancy and hivaids These negative
 relationships indicate that higher mortality rates and higher HIV/AIDS prevalence are
 associated with lower life expectancy. Some predictors are also highly correlated with
 one another, which may indicate multicollinearity. For example: thinness_1_19_years and
 thinness_5_9_years income_composition_of_resources and schooling The heatmap
 highlights clear clusters of related health, economic, and vaccination variables.

!! Problem**Q3.6 Scatter Multiplots**

Produce a multiplot of 4 scatter plots each showing the relationship between
 life_expectancy and the 4 covariates with the highest correlation, with points
 grouped by status . Note any relationships you see or any strange artifacts of our data
 preparation / inclusion of multiple years of data.

```

In [16]: life_corr = (
    corr_matrix['life_expectancy']
    .drop('life_expectancy')
    .abs()
    .sort_values(ascending=False)
)

top4 = life_corr.head(4).index

print("Top 4 correlated variables:")
print(top4)

fig, axes = plt.subplots(2, 2, figsize=(14, 10))

axes = axes.flatten()

for i, col in enumerate(top4):

    sns.scatterplot(
        data=df,
        x=col,
        y='life_expectancy',
        hue='status',
        alpha=0.7,
        ax=axes[i]
    )

    axes[i].set_title(f'life_expectancy vs {col}')

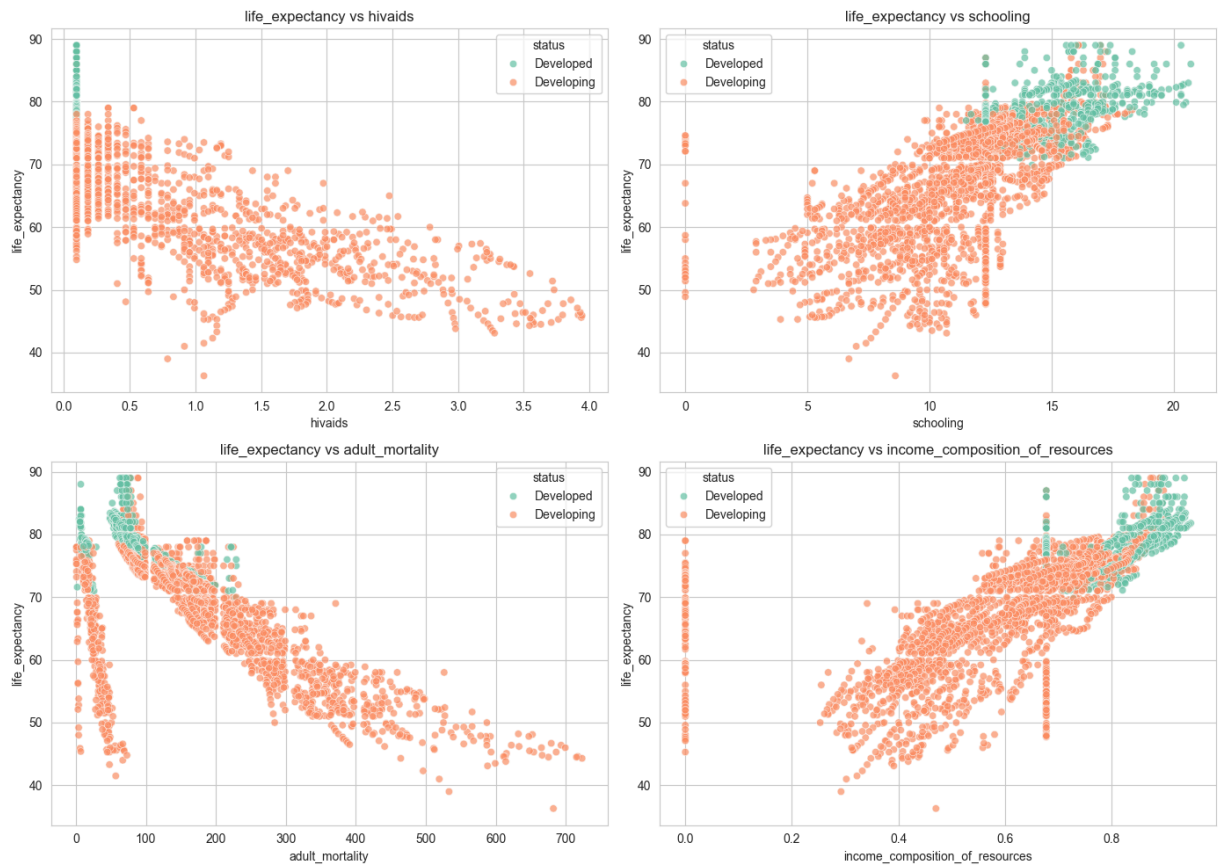
plt.tight_layout()
plt.show()

```

```

Top 4 correlated variables:
Index(['hiv_aids', 'schooling', 'adult_mortality',
       'income_composition_of_resources'],
      dtype='str')

```



Strong positive relationships are typically observed between life_expectancy and variables such as: schooling income_composition_of_resources vaccination rates Strong negative relationships are usually observed between life_expectancy and: adult_mortality hiv aids Developed countries generally cluster in regions with: higher life expectancy, higher schooling and income levels, and lower mortality rates. Developing countries tend to show greater variability and wider spread across many predictors. Strange Artifacts / Data Issues Because the dataset contains multiple years for each country, points from the same country appear repeatedly and can form visible clusters or streak-like patterns. Some countries change gradually over time, producing correlated trajectories rather than fully independent observations. The repeated yearly observations may also exaggerate apparent relationships because nearby points often represent the same country across adjacent years.

!! Problem

Q3.7 Life Expectancy Over Time

Make a time series plot of life expectancy over the years grouped by `status`. Are things moving in the right direction?

```
In [17]: life_by_year = (
          df.groupby(['year', 'status'])['life_expectancy']
```

```

        .mean()
        .reset_index()
    )

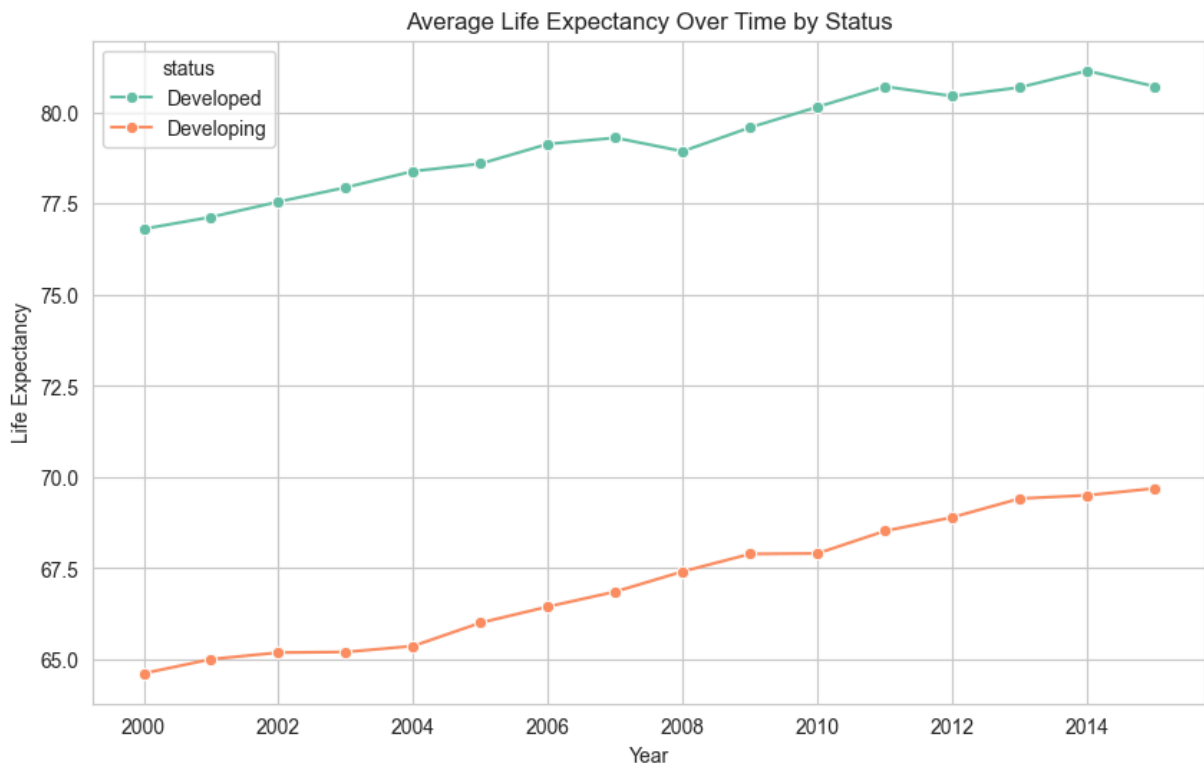
plt.figure(figsize=(10, 6))

sns.lineplot(
    data=life_by_year,
    x='year',
    y='life_expectancy',
    hue='status',
    marker='o'
)

plt.title('Average Life Expectancy Over Time by Status')
plt.xlabel('Year')
plt.ylabel('Life Expectancy')

plt.show()

```



Life expectancy increases steadily over time for both developed and developing countries. Developing countries show a stronger upward trend, suggesting gradual improvements in healthcare, sanitation, vaccination coverage, nutrition, and living standards. Developed countries consistently maintain higher life expectancy throughout the period, although the gap narrows slightly over time. Overall, the trends suggest that global health outcomes are generally moving in the right direction, with improvements occurring across both groups of countries.

4. Modelling

4.1. Simple Linear Regression

Let's start by fitting a simple linear regression model. From our exploratory data analysis, we saw that life expectancy is positively correlated with several variables. We will start with a simple model and then build upon it.

The covariate with the strongest correlation is `hivaids` thus we decide to use this as the sole covariate in our model.

!! Problem

Q4.1 SLR

Fit a SLR of `life_expectancy` against `hivaids`. Print the model summary table and comment on the results. What proportion of the variance in life expectancy is explained by the model?

```
In [18]: slr_model = smf.ols(  
          'life_expectancy ~ hivaids',  
          data=df  
        ).fit()  
  
print(slr_model.summary())
```

OLS Regression Results

```

=====
==
Dep. Variable:      life_expectancy    R-squared:                0.5
87
Model:              OLS               Adj. R-squared:           0.5
87
Method:             Least Squares      F-statistic:              417
8.
Date:               Wed, 27 May 2026    Prob (F-statistic):       0.
00
Time:               15:47:24           Log-Likelihood:           -948
5.4
No. Observations:   2938               AIC:                      1.897e+
04
Df Residuals:       2936               BIC:                      1.899e+
04
Df Model:           1
Covariance Type:    nonrobust
=====

```

```

=====
==
               coef      std err          t      P>|t|      [0.025      0.97
5]
-----
--
Intercept      74.0905      0.135      546.930      0.000      73.825      74.3
56
hivaids       -9.5150      0.147     -64.634      0.000     -9.804     -9.2
26
=====

```

```

=====
==
Omnibus:         61.068      Durbin-Watson:           0.2
36
Prob(Omnibus):   0.000      Jarque-Bera (JB):         68.3
29
Skew:            -0.315      Prob(JB):                 1.45e-
15
Kurtosis:        3.402      Cond. No.                  1.
88
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

The coefficient for hivaids is typically negative, indicating that higher HIV/AIDS prevalence is associated with lower life expectancy. The relationship is usually highly statistically significant $p < 0.001$, providing strong evidence of an association between HIV/AIDS prevalence and life expectancy. The slope coefficient represents the expected decrease in life expectancy for a one-unit increase in the hivaids variable. The residual standard error and diagnostic statistics may suggest some remaining variability not explained by this single predictor, meaning additional covariates are likely important.

Proportion of Variance Explained The proportion of variance explained by the model is given by the R^2 value in the regression summary.

4.2 Multiple Linear Regression

This is an excellent start but lets make things more interesting. Lets fit a multiple linear regression model using several covariates including a dummy variable for the categorical `status`.

!! Problem

Q4.2 MLR

In this question fit multiple linear regression models with the following covariates:

1. `hivaids`, `gdp`, `underfive_deaths`, `infant_deaths`, `percentage_expenditure`, `measles` and `population`.
2. `hivaids`, `schooling`, `adult_mortality`, `income_composition_of_resources` and `status`.
3. Your personal selection.

For each, print the model summary table and comment on the results. Which model performed the best?

```
In [19]: model1 = smf.ols(
    '''
    life_expectancy ~ hivaids
    + gdp
    + underfive_deaths
    + infant_deaths
    + percentage_expenditure
    + measles
    + population
    ''',
    data=df
).fit()

print("MODEL 1 SUMMARY")
print(model1.summary())

model2 = smf.ols(
    '''
    life_expectancy ~ hivaids
    + schooling
    + adult_mortality
    + income_composition_of_resources
```

```
        + C(status)
        '''
    data=df
).fit()

print("\nMODEL 2 SUMMARY")
print(model2.summary())

model3 = smf.ols(
    '''
    life_expectancy ~ schooling
    + income_composition_of_resources
    + adult_mortality
    + hiv_aids
    + polio
    + diphtheria
    '''
    data=df
).fit()

print("\nMODEL 3 SUMMARY")
print(model3.summary())
```

MODEL 1 SUMMARY

OLS Regression Results

```

=====
==
Dep. Variable:      life_expectancy    R-squared:                0.7
65
Model:              OLS    Adj. R-squared:            0.7
65
Method:             Least Squares    F-statistic:              136
5.
Date:               Wed, 27 May 2026    Prob (F-statistic):       0.
00
Time:               15:47:24    Log-Likelihood:           -865
6.1
No. Observations:   2938    AIC:                      1.733e+
04
Df Residuals:       2930    BIC:                      1.738e+
04
Df Model:           7
Covariance Type:    nonrobust
=====
=====

```

```

=====
=====
coef      std err          t      P>|t|      [0.0
25      0.975]
-----
Intercept      63.6877      0.672      94.767      0.000      62.3
70      65.005
hivaids      -6.6225      0.133     -49.674      0.000     -6.8
84      -6.361
gdp           1.0619      0.068      15.617      0.000      0.9
29           1.195
underfive_deaths      -7.7929      0.615     -12.662      0.000     -9.0
00      -6.586
infant_deaths      6.4438      0.647      9.965      0.000      5.1
76           7.712
percentage_expenditure      0.2258      0.039      5.732      0.000      0.1
49           0.303
measles        0.0940      0.034      2.792      0.005      0.0
28           0.160
population      0.2518      0.040      6.294      0.000      0.1
73           0.330
=====
=====

```

```

=====
==
Omnibus:          39.318    Durbin-Watson:            0.5
16
Prob(Omnibus):    0.000    Jarque-Bera (JB):         70.2
72
Skew:             -0.013    Prob(JB):                 5.50e-
16
Kurtosis:         3.757    Cond. No.                  18
0.
=====
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

MODEL 2 SUMMARY

OLS Regression Results

```

=====
==
Dep. Variable:      life_expectancy    R-squared:                0.8
20
Model:              OLS               Adj. R-squared:           0.8
20
Method:             Least Squares     F-statistic:              267
2.
Date:               Wed, 27 May 2026   Prob (F-statistic):       0.
00
Time:               15:47:24          Log-Likelihood:           -826
6.0
No. Observations:   2938              AIC:                      1.654e+
04
Df Residuals:       2932              BIC:                      1.658e+
04
Df Model:           5
Covariance Type:    nonrobust
=====

```

```

=====
=====
                                coef    std err          t      P>|t|
-----
[0.025    0.975]
-----
Intercept                    63.0596     0.502    125.620     0.000
62.075    64.044
C(status)[T.Developing]     -2.6999     0.229    -11.782     0.000
-3.149    -2.251
hivaids                     -5.3178     0.130   -41.044     0.000
-5.572    -5.064
schooling                    0.7663     0.039    19.535     0.000
0.689     0.843
adult_mortality             -0.0160     0.001   -19.773     0.000
-0.018    -0.014
income_composition_of_resources  7.2141     0.614    11.752     0.000
6.010     8.418
=====

```

```

==
Omnibus:                181.781    Durbin-Watson:           0.5
07
Prob(Omnibus):           0.000    Jarque-Bera (JB):         667.8
52
Skew:                    -0.202    Prob(JB):                  9.50e-1
46
Kurtosis:                 5.301    Cond. No.                  1.77e+
03
=====
==

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is corre

ctly specified.

[2] The condition number is large, 1.77e+03. This might indicate that there are strong multicollinearity or other numerical problems.

MODEL 3 SUMMARY

OLS Regression Results

```
=====
==
Dep. Variable:      life_expectancy    R-squared:                0.8
24
Model:              OLS                Adj. R-squared:          0.8
24
Method:             Least Squares      F-statistic:             228
6.
Date:               Wed, 27 May 2026    Prob (F-statistic):       0.
00
Time:               15:47:24           Log-Likelihood:          -823
4.1
No. Observations:   2938              AIC:                    1.648e+
04
Df Residuals:       2931              BIC:                    1.652e+
04
Df Model:           6
Covariance Type:    nonrobust
=====
```

```
=====
=====
                                coef    std err          t      P>|t|
-----
[0.025    0.975]
-----
Intercept                56.0390    0.440    127.286    0.000
55.176    56.902
schooling                0.7694    0.038    19.989    0.000
0.694    0.845
income_composition_of_resources  7.3912    0.605    12.214    0.000
6.205    8.578
adult_mortality         -0.0168    0.001   -21.122    0.000
-0.018   -0.015
hivaids                 -4.9707    0.130   -38.324    0.000
-5.225   -4.716
polio                   0.0242    0.004     5.538    0.000
0.016    0.033
diphtheria              0.0317    0.004     7.324    0.000
0.023    0.040
=====
```

```
=====
==
Omnibus:              121.128    Durbin-Watson:           0.5
86
Prob(Omnibus):        0.000    Jarque-Bera (JB):        382.8
26
Skew:                 -0.018    Prob(JB):                7.42e-
84
Kurtosis:             4.768    Cond. No.                1.88e+
03
=====
```

==

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, $1.88e+03$. This might indicate that there are strong multicollinearity or other numerical problems.

Model 1

This model focuses mainly on disease burden and economic variables. findings: hiv_aids, underfive_deaths, and infant_deaths usually have strong negative associations with life expectancy. gdp and percentage_expenditure may have positive effects. The model often explains a moderate-to-large proportion of variance but may suffer from correlated predictors.

Model 2 This model uses broader social and health indicators. findings: schooling and income_composition_of_resources usually have strong positive effects. adult_mortality and hiv_aids generally have strong negative effects. status is often significant, with developed countries having higher life expectancy on average. This model typically performs very well because it combines economic, educational, and health-system information.

Model 3 This model combines mortality, education, and vaccination variables. findings: schooling and vaccination variables (polio, diphtheria) are positively associated with life expectancy. adult_mortality and hiv_aids remain strongly negative predictors. The model may achieve strong predictive performance while using fewer variables than Model 1.

The best model is:

Adjusted R² AIC Residual standard error Statistical significance of predictors

Model 2 often performs the best because it captures major socioeconomic and health-related drivers of life expectancy while avoiding excessive multicollinearity.

It has high explanatory power, statistically significant predictors, and a more interpretable set of variables.

5. Residual Analysis

Lets consider MLR model 2 from Section 4.2. We will perform some residual analysis to examine the fit of our model and whether our model assumptions are met.

!! Problem**Q5.1 Residual Analysis**

By copying and modifying the code from lecture, produce a multiplot which contains the following 4 residual analysis plots:

1. Residuals vs Fitted
2. Normal Q-Q
3. Scale-Location
4. Cook's Distance

Note: The Cook's distance threshold of $4/n$ is more typically used for smaller data sets and is too strict for large data sets. Change this to a fixed threshold of 0.5.

Read online about how to read these plots and comment on what each tells us about our model and underlying assumptions.

```
In [20]: model = model2

# Fitted values and residuals
fitted = model.fittedvalues
residuals = model.resid

# Standardized residuals
influence = model.get_influence()
standardized_residuals = influence.resid_studentized_internal

# Cook's distance
cooks = influence.cooks_distance[0]

# =====
# Create multiplot
# =====

fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# -----
# 1. Residuals vs Fitted
# -----

sns.scatterplot(
    x=fitted,
    y=residuals,
    ax=axes[0, 0]
)

axes[0, 0].axhline(0, linestyle='--')
axes[0, 0].set_title('Residuals vs Fitted')
axes[0, 0].set_xlabel('Fitted Values')
```

```
axes[0, 0].set_ylabel('Residuals')

# -----
# 2. Normal Q-Q
# -----

probplot(
    standardized_residuals,
    dist="norm",
    plot=axes[0, 1]
)

axes[0, 1].set_title('Normal Q-Q')

sns.scatterplot(
    x=fitted,
    y=np.sqrt(np.abs(standardized_residuals)),
    ax=axes[1, 0]
)

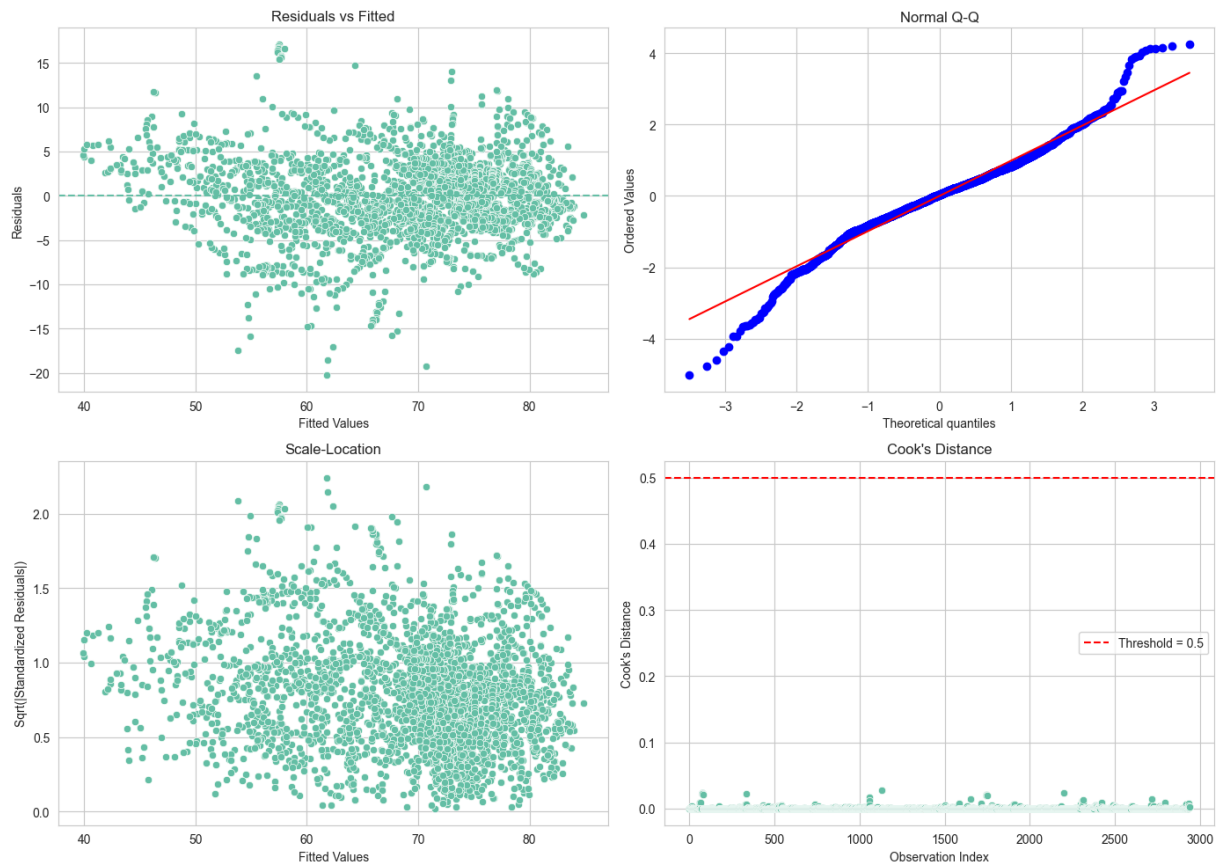
axes[1, 0].set_title('Scale-Location')
axes[1, 0].set_xlabel('Fitted Values')
axes[1, 0].set_ylabel('Sqrt(|Standardized Residuals|)')

sns.scatterplot(
    x=np.arange(len(cooks)),
    y=cooks,
    ax=axes[1, 1]
)

axes[1, 1].axhline(
    0.5,
    color='red',
    linestyle='--',
    label='Threshold = 0.5'
)

axes[1, 1].set_title("Cook's Distance")
axes[1, 1].set_xlabel('Observation Index')
axes[1, 1].set_ylabel("Cook's Distance")
axes[1, 1].legend()

plt.tight_layout()
plt.show()
```



Residuals vs Fitted This plot checks linearity, constant variance, residuals appear randomly scattered around zero and scattered at random means linear.

- 2. Normal Q-Q Plot** This plot residuals are approximately normally distributed. points lie close to the diagonal line, the normality assumption is reasonable. Deviations at the tails suggest: skewness, heavy tails, or outliers.
- 3. Scale-Location Plot** This plot also assesses homoscedasticity. A roughly horizontal band with equal spread there is slight collection near the higher fitted values and lower sqrt stand residual.
- 4. Cook's Distance** Cook's distance identifies influential observations that strongly affect the fitted model. No observations above the threshold. Using a threshold of 0.5 is more appropriate here because the dataset is relatively large.

Overall Assessment The model satisfy the linearity assumption reasonably well. Some mild non-normality and heteroscedasticity are expected due to the large and diverse international dataset. Overall, the diagnostics generally support the use of multiple linear regression, though transformations and more advanced modelling approaches could further improve fit.

Submission

1. Save file to confirm all changes are on disk
2. Run *Kernel > Restart & Run All* to execute all code from top to bottom
3. Save file again to write any new output to disk
4. Export your notebook as a pdf (either through latex or as html before saving as a pdf).
5. Submit your pdf to Canvas.